INSTITUTO DE FÍSICA DA USP

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QUANTUM FIELD THEORY I

Homework Set 1

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Quantum Field Theory I - Homework Set 1

Question 1

Let $J^{\mu\nu}$ be the generators of the Lorentz group.

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(a) Show that the Lie algebra of the Lorentz group is

$$i[J^{\alpha\beta}, J^{\rho\sigma}] = g^{\sigma\beta}J^{\rho\alpha} + g^{\rho\alpha}J^{\sigma\beta} + g^{\rho\beta}J^{\alpha\sigma} + g^{\alpha\sigma}J^{\beta\rho}.$$

(b) Obtain the irreducible representations of the Lorentz group.

(a) The vector representation of Lorentz group generators is

$$[J^{\mu\nu}]^{\rho}{}_{\sigma} = i\left(g^{\mu\rho}\delta^{\nu}_{\sigma} - g^{\nu\rho}\delta^{\mu}_{\sigma}\right) \tag{1}$$

since this is the usual vector representation of the Lorentz group, which acts on four-vectors x^ρ as $x^\rho \to x^\rho + \frac{1}{2}\omega_{\mu\nu}\left(J^{\mu\nu}\right)^{\rho}{}_{\sigma}x^\sigma$, reproducing the infinitesimal Lorentz transformation $x'^\rho = x^\rho + \omega^\rho{}_\sigma x^\sigma$. Then, the commutator can be written as:

↳ missing $-i$

↳ you should have shown this!

$$\left[J^{\alpha\beta}, J^{\rho\sigma}\right]^{\mu}{}_{\nu} = \left[J^{\alpha\beta}\right]^{\mu}{}_{\gamma}\left[J^{\rho\sigma}\right]^{\gamma}{}_{\nu} - \left[J^{\rho\sigma}\right]^{\mu}{}_{\gamma}\left[J^{\alpha\beta}\right]^{\gamma}{}_{\nu}$$

$$\begin{aligned}
&= -\left(g^{\alpha\mu}\delta^{\beta}_{\gamma} - g^{\beta\mu}\delta^{\alpha}_{\gamma}\right)\left(g^{\rho\gamma}\delta^{\sigma}_{\nu} - g^{\sigma\gamma}\delta^{\rho}_{\nu}\right) + \left(g^{\rho\mu}\delta^{\sigma}_{\gamma} - g^{\sigma\mu}\delta^{\rho}_{\gamma}\right)\left(g^{\alpha\gamma}\delta^{\beta}_{\nu} - g^{\beta\gamma}\delta^{\alpha}_{\nu}\right)\\
&= -g^{\alpha\mu}\delta^{\beta}_{\gamma}g^{\rho\gamma}\delta^{\sigma}_{\nu} + g^{\alpha\mu}\delta^{\beta}_{\gamma}g^{\sigma\gamma}\delta^{\rho}_{\nu} + g^{\beta\mu}\delta^{\alpha}_{\gamma}g^{\rho\gamma}\delta^{\sigma}_{\nu} - g^{\beta\mu}\delta^{\alpha}_{\gamma}g^{\sigma\gamma}\delta^{\rho}_{\nu}\\
&\quad + g^{\rho\mu}\delta^{\sigma}_{\gamma}g^{\alpha\gamma}\delta^{\beta}_{\nu} - g^{\rho\mu}\delta^{\sigma}_{\gamma}g^{\beta\gamma}\delta^{\alpha}_{\nu} - g^{\sigma\mu}\delta^{\rho}_{\gamma}g^{\alpha\gamma}\delta^{\beta}_{\nu} + g^{\sigma\mu}\delta^{\rho}_{\gamma}g^{\beta\gamma}\delta^{\alpha}_{\nu}.
\end{aligned}$$

Using $g^{\alpha\beta}\delta^{\gamma}_{\beta} = g^{\alpha\gamma}$, we can rewrite and rearrange this expression as

$$\begin{aligned}
&= -g^{\alpha\mu}g^{\rho\beta}\delta^{\sigma}_{\nu} + g^{\alpha\mu}g^{\sigma\beta}\delta^{\rho}_{\nu} + g^{\beta\mu}g^{\rho\alpha}\delta^{\sigma}_{\nu} - g^{\beta\mu}g^{\sigma\alpha}\delta^{\rho}_{\nu} + g^{\rho\mu}g^{\alpha\sigma}\delta^{\beta}_{\nu} - g^{\rho\mu}g^{\beta\sigma}\delta^{\alpha}_{\nu} - g^{\sigma\mu}g^{\alpha\rho}\delta^{\beta}_{\nu} + g^{\sigma\mu}g^{\beta\rho}\delta^{\alpha}_{\nu}\\
&= g^{\sigma\beta}\left(g^{\alpha\mu}\delta^{\rho}_{\nu} - g^{\rho\mu}\delta^{\alpha}_{\nu}\right) + g^{\rho\alpha}\left(g^{\beta\mu}\delta^{\sigma}_{\nu} - g^{\sigma\mu}\delta^{\beta}_{\nu}\right) + g^{\rho\beta}\left(g^{\sigma\mu}\delta^{\alpha}_{\nu} - g^{\alpha\mu}\delta^{\sigma}_{\nu}\right) + g^{\alpha\sigma}\left(g^{\rho\mu}\delta^{\beta}_{\nu} - g^{\beta\mu}\delta^{\rho}_{\nu}\right)
\end{aligned}$$

$$\Rightarrow i\left[J^{\alpha\beta}, J^{\rho\sigma}\right]^{\mu}{}_{\nu} = g^{\sigma\beta}[J^{\alpha\rho}]^{\mu}{}_{\nu} + g^{\rho\alpha}[J^{\beta\sigma}]^{\mu}{}_{\nu} + g^{\rho\beta}[J^{\sigma\alpha}]^{\mu}{}_{\nu} + g^{\alpha\sigma}[J^{\rho\beta}]^{\mu}{}_{\nu}.$$

Alternatively, knowing that $J^{\alpha\beta} = -J^{\beta\alpha}$ (check eq. (1)), we can write this result as

$$\boxed{i[J^{\alpha\beta}, J^{\rho\sigma}] = -g^{\sigma\beta}J^{\rho\alpha} - g^{\rho\alpha}J^{\sigma\beta} - g^{\rho\beta}J^{\alpha\sigma} - g^{\alpha\sigma}J^{\beta\rho}} \tag{2}$$

or, more commonly expressed as (using the anti symmetry property of the generators):

$$[J^{\alpha\beta}, J^{\rho\sigma}] = ig^{\rho\beta}J^{\alpha\sigma} - ig^{\sigma\beta}J^{\alpha\rho} - ig^{\alpha\rho}J^{\beta\sigma} + ig^{\alpha\sigma}J^{\beta\rho}. \tag{3}$$ OK

(Even though the signs differ from the expression given in the problem, they are compatible with equation (2.41) from the textbook *Field Theory: A Modern Primer*, by Pierre Ramond.) → matter of metric conventions & passive vs. active transformations

(b) We can split the Lorentz group generators into 3 rotations and 3 boosts:

$$J^k = \frac{1}{2}\epsilon^{ijk}J^{jk} \text{ (rotations)}, \tag{4}$$

watch out spatial indices if you using $g = + - - -$

$$K^j = J^{j0} \text{ (boosts)}, \tag{5}$$

where ϵ is the Levi-Civita symbol. For the rotations, we compute

$$[J_i, J_j] = \frac{1}{4}\epsilon_{iab}\epsilon_{jcd}[J^{ab}, J^{cd}] \tag{6}$$

$$= \frac{i}{4}\epsilon_{iab}\epsilon_{jcd}\left(g^{bc}J^{ad} - g^{ac}J^{bd} - g^{bd}J^{ac} + g^{ad}J^{bc}\right). \tag{7}$$

Since the indices are purely spatial, $g^{ab} = -\delta^{ab}$, and performing the contractions,

$$[J_i, J_j] = i\epsilon_{ijk}J_k. \tag{8}$$

you should have shown the intermediate steps....

For the boosts,

$$[K_i, K_j] = [J^{i0}, J^{j0}] \tag{9}$$

$$= i\left(g^{0j}J^{i0} - g^{ij}J^{00} - g^{00}J^{ij} + g^{i0}J^{0j}\right). \tag{10}$$

Using $g^{00} = +1$, $g^{i0} = g^{0j} = 0$, and $J^{00} = 0$,

$$[K_i, K_j] = -iJ^{ij} = -i\epsilon_{ijk}J_k. \tag{11}$$

you should have shown $J^{ij} = \epsilon^{ijk} J_k$

For the mixed commutator,

$$[J_i, K_j] = \frac{1}{2}\epsilon_{iab}[J^{ab}, J^{j0}] \tag{12}$$

$$= \frac{i}{2}\epsilon_{iab}\left(g^{bj}J^{a0} - g^{aj}J^{b0}\right). \tag{13}$$

Using $g^{aj} = -\delta^{aj}$,

$$[J_i, K_j] = i\epsilon_{ijk}K_k. \tag{14}$$

The three commutation relations are therefore

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \tag{15}$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k, \tag{16}$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k. \tag{17}$$

ok

From these, we can create linear combinations:

$$J_k^{\pm} = \frac{1}{2}(J_k \pm iK_k), \tag{18}$$

so that $J_k = J_k^+ + J_k^-$ and $K_k = -i(J_k^+ - J_k^-)$. Computing $[J_i^+, J_j^+]$ explicitly,

$$[J_i^+, J_j^+] = \frac{1}{4}[J_i + iK_i,\, J_j + iK_j] \tag{19}$$

$$= \frac{1}{4}\Big([J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] - [K_i, K_j]\Big) \tag{20}$$

$$= \frac{1}{4}\Big(i\epsilon_{ijk}J_k - \epsilon_{ijk}K_k - \epsilon_{ijk}K_k + i\epsilon_{ijk}J_k\Big) \tag{21}$$

$$= \frac{1}{4}\Big(2i\epsilon_{ijk}J_k - 2\epsilon_{ijk}K_k\Big) \tag{22}$$

$$= i\epsilon_{ijk}\frac{1}{2}(J_k + iK_k) \tag{23}$$

$$= i\epsilon_{ijk}\,J_k^+. \tag{24}$$

ok

Similarly, we obtain

you should have shown

$$[J_i^-, J_j^-] = i\epsilon_{ijk}\,J_k^-, \tag{25}$$

$$[J_i^+, J_j^-] = 0. \tag{26}$$

$so(1,3) \cong so(3,1)$ are non-compact algebras, while $su(2) \oplus su(2)$ is a compact algebra: $so(3,1) \ncong su(2) \oplus su(2)$. The correct statement is derived in the solutions

Therefore, we can conclude that the algebra of the Lorentz group $SO(1,3)$ is isomorphic to a direct sum of two $su(2)$ algebras:

$$so(3,1) \cong su(2) \oplus su(2). \tag{27}$$

→ wrong!

But from Quantum Mechanics, we know that the irreducible representations of $SU(2)$ are labeled by integer and semi-integer numbers ($j = 0, \frac{1}{2}, 1, \ldots$). Therefore, the irreducible representations of the Lorentz group are given by a pair (j_1, j_2) where $j_1, j_2 = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$.

OK

$so(1,3)|_{\mathbb{C}} \cong su(2)|_{\mathbb{C}} \oplus su(2)|_{\mathbb{C}}$

Question 2

Consider the four-vector electromagnetic potential

$$A^\mu = (\Phi, \vec{A}),$$

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where Φ ($\vec{A}$) is the scalar (vector) potential. In the absence of sources the electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

(a) Obtain the classical equations of motion.

(b) Obtain the energy-momentum tensor that is the conserved current when we consider the symmetry $x^\mu \to x'^\mu = x^\mu + a^\mu$, where a^μ is a constant.

(a) First, let us derive the Euler-Lagrange equation. For a continuous system, the action is defined as

$$S[\varphi] = \int d^4x \, \mathcal{L}(\varphi, \partial_\mu \varphi). \tag{28}$$

We know, from the principle of least action, that $\delta S = 0$. Consider the following transformation:

$$\varphi \to \varphi + \delta\varphi,$$

$$\partial_\mu \phi \to \partial_\mu \phi + \partial_\mu \delta\varphi.$$

Explicitly computing the variation in the action:

$$\delta S = \int d^4x \, \left[\mathcal{L}(\varphi + \delta\varphi, \partial_\mu \varphi + \partial_\mu \delta\varphi) - \mathcal{L}(\varphi, \partial_\mu \varphi)\right]. \tag{29}$$

Taylor expanding up to first order:

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \varphi} \delta\varphi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu \delta\varphi\right]. \tag{30}$$

Integrating the second term by parts and using boundary conditions (fixed endpoints):

→ actually the boundary conditions only fix $\delta\phi = 0$ at the temporal boundary. At the spatial boundary you get other set of E.o.M.

$$\delta S = \int d^4x \left[\delta\varphi \frac{\partial \mathcal{L}}{\partial \varphi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \delta\varphi\right) = 0\right], \tag{31}$$

and we obtain the Euler-Lagrange equation for continuous systems:

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)}\right) = 0. \tag{32}$$

OK

Now, in order to obtain the equations of motion for any system, we must simply apply eq. (32) to a given Lagrangian. In this case, we have the electromagnetic Lagrangian (no sources):

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \tag{33}$$

where $F^{\mu\nu}$ is the electromagnetic tensor, defined as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. From eq. (32), we'll have:

$$\cancelto{0}{\frac{\partial \mathcal{L}}{\partial A_\nu}} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) = 0 \tag{34}$$

$$\implies \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) = 0. \tag{35}$$

Computing the derivative of the Lagrangian:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{4} \frac{\partial}{\partial(\partial_\mu A_\nu)} (F_{\alpha\beta} F^{\alpha\beta}) = -\frac{1}{4} \left[\frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} F^{\alpha\beta} + F_{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} \right]. \tag{36}$$

Notice that:

$$\frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} = \frac{\partial}{\partial(\partial_\mu A_\nu)} [\partial_\alpha A_\beta - \partial_\beta A_\alpha] = \delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha, \tag{37}$$

where δ^α_β is the Kronecker delta. Then:

$$\frac{\partial F^{\alpha\beta}}{\partial(\partial_\mu A_\nu)} = g^{\sigma\alpha} g^{\rho\beta} \frac{\partial F_{\sigma\rho}}{\partial(\partial_\mu A_\nu)} = g^{\sigma\alpha} g^{\rho\beta} (\delta^\mu_\sigma \delta^\nu_\rho - \delta^\mu_\rho \delta^\nu_\sigma) = g^{\alpha\mu} g^{\beta\nu} - g^{\nu\alpha} g^{\mu\beta}. \tag{38}$$

Substituting these results in eq. (36):

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{4} \left[(\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha) F^{\alpha\beta} + F_{\alpha\beta} (g^{\alpha\mu} g^{\beta\nu} - g^{\nu\alpha} g^{\mu\beta}) \right] \tag{39}$$

$$= -\frac{1}{4} (2F^{\mu\nu} - 2F^{\nu\mu}) \tag{40}$$

$$= -\frac{1}{4} (4F^{\mu\nu}), \tag{41}$$

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where in the last step we used $F^{\mu\nu} = -F^{\nu\mu}$. Finally, plugging this into eq. (35), we obtain the equations of motion:

$$\boxed{\partial_\mu F^{\mu\nu} = 0} \tag{42}$$

This expression represents the two inhomogeneous Maxwell's equations (Gauss's law and Ampère's law) in the absence of sources.

(b) For a continuous symmetry transformation:

$$x \to x',$$

$$\varphi(x) \to \varphi'(x').$$

First, we define

$$\delta\varphi(x) = \varphi'(x) - \varphi(x). \tag{43}$$

The variation in the action can be explicitly evaluating the expression from eq. (30). In order to keep the action invariant, the Lagrangian must only differ from the original Lagrangian by a total derivative:

$$\delta S = \int d^4x \, \partial_\mu \Lambda^\mu = 0. \tag{44}$$

But we could also apply the Euler-Lagrange equation (eq. (32)) in eq. (30) to get

$$\delta S = \int d^4x \left[\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \right) \delta\varphi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \partial_\mu \delta\varphi \right] = \int d^4x \, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \delta\varphi \right). \tag{45}$$

Combining both results:

$$\partial_\mu \Lambda^\mu = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \delta\varphi \right) \implies \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \delta\varphi - \Lambda^\mu \right) = 0. \tag{46}$$

This result is known as Noether's Theorem, and it gives us the conserved current j^μ of a system:

$$j^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \delta\varphi - \Lambda^\mu. \tag{47}$$

ok

Now we have everything we need to find the conserved current for the electromagnetic Lagrangian. Consider the transformation:

$$x^\mu \to x'^\mu = x^\mu + a^\mu,$$
$$\varphi'(x^\mu + a^\mu) = \varphi(x^\mu).$$

The variation in the field:

$$\delta\varphi(x^\mu) = \varphi'(x^\mu) - \varphi(x^\mu) = \varphi(x^\mu - a^\mu) - \varphi(x^\mu), \tag{48}$$

and Taylor expanding up to first order:

$$\delta\varphi(x^\mu) = \varphi(x^\mu) - a^\mu \partial_\mu \varphi(x^\mu) - \varphi(x^\mu) = -a^\mu \partial_\mu \varphi(x^\mu). \tag{49}$$

ok

Going back to the expression of the variation of the action (eq. (30)), we'll have

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \varphi} (-a^\rho \partial_\rho \varphi) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \partial_\mu (-a^\rho \partial_\rho \varphi) \right]. \tag{50}$$

Notice that it can be rewritten as a total derivative:

$$\int d^4x \, \partial_\rho (-a^\rho \mathcal{L}). \tag{51}$$

Comparing the expression above with eq. (44), we identify

$$\Lambda^\mu = -a^\mu \mathcal{L}. \tag{52}$$

Applying the Noether's Theorem (eq. (46)), we can finally find the conserved current for such transformation:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} [-a^\rho \partial_\rho \varphi] + a^\mu \mathcal{L} \right) = 0 \tag{53}$$

$$= \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} a^\rho \partial_\rho \varphi - \mathcal{L} a^\rho \delta^\mu_\rho \right) = 0 \tag{54}$$

$$= a^\rho \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \partial_\rho \varphi - \mathcal{L} \delta^\mu_\rho \right) = 0. \tag{55}$$

Thus, the conserved current is (calling it ${T^\mu}_\rho$ instead of ${j^\mu}_\rho$):

$${T^\mu}_\rho = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \partial_\rho \varphi - \mathcal{L} \delta^\mu_\rho. \tag{56}$$

ok

Raising the index, we obtain the canonical energy-momentum tensor:

$$T^{\mu\rho} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \partial^\rho \varphi - \mathcal{L} g^{\mu\rho}. \tag{57}$$

Lastly, we can particularize this result for the electromagnetic Lagrangian as follow:

$$T^{\mu}{}_{\rho} = \frac{\partial}{\partial(\partial_\mu A_\nu)}\left(-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\right)\partial_\rho A_\nu + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\delta^{\mu}_{\rho} \tag{58}$$

$$= -\frac{1}{4}\left(\frac{\partial F_{\alpha\beta}}{\partial(\partial\mu A_\nu)}F^{\alpha\beta} + F_{\alpha\beta}\frac{\partial F^{\alpha\beta}}{\partial(\partial\mu A_\nu)}\right)\partial_\mu A_\nu + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\delta^{\mu}_{\rho}. \tag{59}$$

Using eq. (37) and eq. (38), we obtain

$$T^{\mu}{}_{\rho} = -F^{\mu\nu}\partial_\rho A_\nu + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\delta^{\mu}_{\rho}, \tag{60}$$

or, with both indices raised:

$$\boxed{T^{\mu\rho} = -F^{\mu\nu}\partial^\rho A_\nu + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}g^{\mu\rho}} \tag{61}$$

ok

It's easy to see that this tensor is not symmetric, though. To make it a symmetric tensor, we add total derivative term:

$$\partial_\nu K^{\nu\mu\rho} = \underbrace{\cancel{(\partial_\nu F^{\mu\nu})A^\rho}}_{0} + F^{\mu\nu}(\partial_\nu A^\rho), \tag{62}$$

where the first term cancels to zero because of the equation of motion (item [a]). Therefore,

$$\hat{T}^{\mu\rho} = F^{\mu\nu}(\partial_\nu A^\rho - \partial^\rho A_\nu) + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}g^{\mu\rho}, \tag{63}$$

and we obtain the symmetrized version of the energy-momentum tensor:

$$\boxed{\hat{T}^{\mu\rho} = F^{\mu\nu}F_\nu{}^\rho + \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}g^{\mu\rho}} \tag{64}$$

ok

Question 3

Consider a real scalar field ϕ whose Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{\lambda}{4!}\phi^4.$$

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A scale transformation is defined as

$$x \to x' = e^\epsilon x, \qquad \phi(x) \to \phi'(x') = e^{-d_\phi \epsilon}\phi(x),$$

where $d_\phi = 1$ is the dimension of the field ϕ.

Is this transformation a symmetry? If so, what is the conserved current?

Let's proceed the same way we did in the previous question. First, we compute the variation in the field.

$$\delta\varphi(x) = \varphi'(x) - \varphi(x) = e^{-\epsilon}\varphi(e^{-\epsilon}x) - \varphi(x). \tag{65}$$

Taylor expanding the first term:

$$e^{-\epsilon}\varphi(e^{-\epsilon}x) = (1-\epsilon)\varphi(x-\epsilon x) = \varphi(x-\epsilon x) - \epsilon\varphi(x-\epsilon x) = \varphi(x) - \epsilon x^\mu\partial_\mu\varphi(x) - \epsilon\varphi(x). \tag{66}$$

Then, we'll have

$$\delta\varphi = -\epsilon(\varphi + x^\mu\partial_\mu\varphi). \tag{67}$$

ok

Now we can compute the variation in the action:

$$\delta S = \int d^4x\left[\frac{\partial\mathcal{L}}{\partial\varphi}\delta\varphi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\delta\varphi\right] \equiv \int d^4x\,\delta\mathcal{L}. \tag{68}$$

Explicitly writing $\delta\mathcal{L}$:

$$\delta\mathcal{L} = \left[\frac{\partial\mathcal{L}}{\partial\varphi}(-\epsilon[\varphi + x^\nu\partial_\nu\varphi]) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(-\epsilon[\varphi + x^\nu\partial_\nu\varphi])\right]. \tag{69}$$

The derivative in the first term can be rewritten as

$$\begin{aligned} -\epsilon\partial_\mu(\varphi + x^\nu\partial_\nu\varphi) &= -\epsilon\left[\partial_\mu\varphi + (\partial_\mu x^\nu)\partial_\nu\varphi + x^\nu\partial_\mu(\partial_\nu\varphi)\right] & (70)\\ &= -\epsilon[2\partial_\mu\varphi + x^\nu\partial_\mu(\partial_\nu\varphi)]. & (71)\end{aligned}$$

Therefore,

$$\begin{aligned} \delta\mathcal{L} &= -\epsilon\left[x^\nu\left(\frac{\partial\mathcal{L}}{\partial\varphi}\partial_\nu\varphi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}\partial_\mu(\partial_\nu\varphi)\right) + \frac{\partial\mathcal{L}}{\partial\varphi}\varphi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}2\partial_\mu\varphi\right] & (72)\\ &= -\epsilon\left[x^\nu\partial_\nu\mathcal{L} + \frac{\partial\mathcal{L}}{\partial\varphi}\varphi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}2\partial_\mu\varphi\right]. & (73)\end{aligned}$$

Notice that

$$\partial_\nu(x^\nu\mathcal{L}) = (\partial_\nu x^\nu)\mathcal{L} + x^\nu\partial_\nu\mathcal{L} = 4\mathcal{L} + \S^\nu\partial_\nu\mathcal{L} \implies x^\nu\partial_\nu\mathcal{L} = \partial_\nu(x^\nu\mathcal{L}) - 4\mathcal{L}, \tag{74}$$

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and we end up with the following expression for the variation in the Lagrangian:

$$\delta\mathcal{L} = -\epsilon\left[\partial_\nu(x^\nu\mathcal{L}) + \frac{\partial\mathcal{L}}{\partial\varphi}\varphi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}2\partial_\mu\varphi - 4\mathcal{L}\right]. \tag{75}$$

If we substitute the Lagrangian of the scalar field, the last three terms will equal to zero:

$$\begin{aligned} &\implies -\frac{\lambda}{3!}\varphi^4 + \frac{\partial}{\partial(\partial_\mu\varphi)}(\partial_\alpha\varphi\partial^\alpha\varphi)\partial_\mu\varphi - 4\mathcal{L} & (76)\\ &= -\frac{\lambda}{3!}\varphi^4 + \left[\frac{\partial}{\partial(\partial_\mu\varphi)}(\partial_\alpha\varphi)\partial^\alpha\varphi + \partial_\alpha\varphi\frac{\partial}{\partial(\partial_\mu\varphi)}(\partial^\alpha\varphi)\right]\partial_\mu\varphi - 4\mathcal{L} & (77)\\ &= -\frac{\lambda}{3!}\varphi^4 + [\delta^\mu_\alpha\partial^\alpha\varphi + \partial_\alpha\varphi g^{\mu\alpha}]\partial_\mu\varphi - 4\mathcal{L} & (78)\\ &= -\frac{\lambda}{3!}\varphi^4 + 2\partial^\mu\varphi\partial_\mu\varphi - 4\mathcal{L} & (79)\\ &= 0. & (80)\end{aligned}$$

Therefore, the variation in the lagrangian is simply

$$\delta\mathcal{L} = -\epsilon\partial_\nu(x^\nu\mathcal{L}), \tag{81}$$

and since the lagrangian differs from the original up to a total derivative, the action doesn't change, the equations of motion remain unchanged, and we can say that this transformation is a symmetry.

Now we just apply the Noether's Theorem to find the conserved current:

Look. but there where simple ways of getting here

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}\delta\varphi - \Lambda^\mu, \tag{82}$$

where $\Lambda^\mu = -\epsilon x^\mu\mathcal{L}$. Then

$$j^\mu = \frac{\partial}{\partial(\partial_\mu\varphi)}\left[\frac{1}{2}\partial_\alpha\varphi\partial^\alpha\varphi\right]\delta\varphi + \epsilon x^\mu\mathcal{L} = \partial^\mu\varphi[-\epsilon(\varphi + x^\nu\partial_\nu\varphi)] + \epsilon x^\mu\mathcal{L}. \tag{83}$$

Since ϵ is arbitrary, we can express the conserved current as (up to an overall factor):

$$\boxed{j^\mu = [\partial^\mu\varphi(x^\nu\partial_\nu\varphi + \varphi) - x^\mu\mathcal{L}]} \tag{84}$$

or

where $\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{\lambda}{4!}\phi^4$.

Question 4

Consider a Lorentz transformation

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}.$$

2/2

An infinitesimal Lorentz transformation takes the form

$$x'^{\mu} = x^{\mu} + \delta\omega^{\mu\nu} x_{\nu}.$$

(a) Show that $\delta\omega^{\mu\nu} = -\delta\omega^{\nu\mu}$.

(b) For a rotation, what is the form of $\delta\omega^{\mu\nu}$?

(c) For a scalar field $\phi(x)$, obtain the conserved current associated to a Lorentz transformation.

(a) We know that

$$g_{\mu\nu} = g_{\alpha\beta} \Lambda^{\alpha}{}_{\mu} \Lambda^{\beta}{}_{\nu}. \tag{85}$$

For a infinitesimal transformation, $\Lambda^{\mu}{}_{\nu} = \delta^{\mu}_{\nu} + \delta\omega^{\mu}{}_{\nu}$, where $\delta\omega^{\mu}{}_{\nu} \ll 1$. From eq. (85),

$$\begin{aligned} g_{\mu\nu} &= g_{\alpha\beta}(\delta^{\alpha}_{\mu} + \delta\omega^{\alpha}{}_{\mu})(\delta^{\beta}_{\nu} + \delta\omega^{\beta}{}_{\nu}) && (86)\\ &= g_{\alpha\beta}(\delta^{\alpha}_{\mu}\delta^{\beta}_{\nu} + \delta^{\alpha}_{\mu}\delta\omega^{\beta}{}_{\nu} + \delta\omega^{\alpha}{}_{\mu}\delta^{\beta}_{\nu}) && (87)\\ &= g_{\mu\nu} + \delta\omega_{\mu\nu} + \delta\omega_{\nu\mu} && (88)\\ \implies & \boxed{\delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu}} && (89) \end{aligned}$$

ok

(b) From the previous item, we already know that the matrix is anti-symmetric. For a rotation, $\delta\omega_{0i} = 0$ (no boosts) and, in 3D, a rotation matrix can be written as $\delta\omega_{ij} = \epsilon_{ijk}\delta\theta_k$, where $\delta\theta_k$ are the infinitesimal angles. The most generic matrix is then

$$\delta\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \theta_3 & -\theta_2 \\ 0 & -\theta_3 & 0 & \theta_1 \\ 0 & \theta_2 & -\theta_1 & 0 \end{pmatrix}. \tag{90}$$

For a rotation around the z-axis, one can write

$$\delta\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \theta & 0 \\ 0 & -\theta & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{91}$$

ok

(c) We are working with a transformation similar to the one we have worked on the second question, so we can take advantage of that (just changing $a^{\mu} \to \delta\omega^{\mu\nu}x_{\nu}$). Comparing to eq. (30) and eq. (52), we'll have

$$\delta\varphi = -\delta\omega^{\mu}{}_{\nu} x^{\nu} \partial_{\mu}\varphi, \tag{92}$$

ok

and

$$\Lambda^{\mu} = -\delta\omega^{\mu}{}_{\nu} x^{\nu} \mathcal{L}. \tag{93}$$

→ requires also that $\partial_\mu(\omega^\mu{}_\nu x^\nu) = 0$

Now we apply the Noether's Theorem to find the conserved current:

$$\partial_{\mu}\left(\frac{\partial\mathcal{L}}{\partial(\partial_{\mu}\varphi)}\delta\varphi - \Lambda^{\mu}\right) = 0 \tag{94}$$

$$\implies \partial_{\mu}\left(\frac{\partial\mathcal{L}}{\partial(\partial_{\mu}\varphi)}[-\delta\omega^{\rho}{}_{\nu}x^{\nu}\partial_{\rho}\varphi] + \delta\omega^{\mu}{}_{\nu}x^{\nu}\mathcal{L}\right) = 0. \tag{95}$$

The conserved current:

$$j^\mu = -\delta\omega^\rho{}_\nu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} x^\nu \partial_\rho\varphi + \delta\omega^\mu{}_\nu x^\nu \mathcal{L}. \tag{96}$$

We can rewrite the first term as

$$-\delta\omega^\alpha{}_\beta \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} x^\beta \partial_\alpha\varphi = -g^{\alpha\gamma}\delta\omega_{\gamma\beta} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} x^\beta \partial_\alpha\varphi = -\delta\omega_{\gamma\beta} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} x^\beta \partial^\gamma\varphi, \tag{97}$$

and the second term:

$$\delta\omega^\mu{}_\nu x^\nu \mathcal{L} = g^{\mu\alpha}\delta\omega_{\alpha\beta} x^\beta \mathcal{L}, \tag{98}$$

and we can rewrite the current as

$$j^\mu = \delta\omega_{\alpha\beta}\left(-x^\beta \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} \partial^\alpha\varphi + g^{\mu\alpha} x^\beta \mathcal{L}\right) \equiv \delta\omega_{\alpha\beta} A^{\mu\alpha\beta}. \tag{99}$$

Since $\delta\omega_{\alpha\beta}$ is anti-symmetric, we can use the following identity:

$$\delta\omega_{\alpha\beta} A^{\alpha\beta} = \frac{1}{2}\delta\omega_{\alpha\beta}(A^{\alpha\beta} - A^{\beta\alpha}), \tag{100}$$

allowing us to, once again, rewrite the current as

$$j^\mu = \frac{1}{2}\delta\omega_{\alpha\beta}\left(-x^\beta \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} \partial^\alpha\varphi + g^{\mu\alpha} x^\beta \mathcal{L} + x^\alpha \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} \partial^\beta\varphi - g^{\mu\beta} x^\alpha \mathcal{L}\right). \tag{101}$$

Identifying the energy momentum tensor (eq. (57)), we can define

$$M^{\mu\alpha\beta} \equiv x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha} \tag{102}$$

→ very good

and finally write the conserved current as

the conserved current is $M^{\mu\alpha\beta}$. It cannot depend upon an arbitrary parameter as ω

$$\boxed{j^\mu = \frac{1}{2}\delta\omega_{\alpha\beta} M^{\mu\alpha\beta}} \tag{103}$$

Question 5

Consider the vector field A^μ associated to the electromagnetic field. Knowing that

$$A'^\mu(x') = \Lambda^\mu_{\ \nu} A^\nu(x),$$

2/2

obtain the conserved quantities due to the Lorentz invariance of the theory. Use the Lagrangian density given in the first problem.

First, let us calculate the variation in the field:

$$A^\mu(x) \to A'^\mu(x') = \Lambda^\mu{}_\nu A^\nu(x),$$

therefore

$$\delta A(x) = A'(x) - A(x) \tag{104}$$

$$\delta A(x) = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x) - A(x). \tag{105}$$

The infinitesimal transformations:

$$\delta A^\mu(x) = (\delta^\mu_\nu + \omega^\mu{}_\nu) A^\nu(x^\rho - \omega^\rho{}_\nu x^\nu) - A^\mu(x). \tag{106}$$

Taylor expanding:

$$\delta A^\mu(x) = (\delta^\mu_\nu + \omega^\mu{}_\nu)[A^\nu(x) - \omega^\rho{}_\nu x^\nu \partial_\rho A^\nu] - A^\mu(x) \tag{107}$$
$$\delta A^\mu(x) = \omega^\mu{}_\nu A^\nu - \omega^\rho{}_\nu x^\nu \partial_\rho A^\mu. \tag{108}$$

or

Now, to obtain the variation in the Lagrangian, we can use the fact that, by definition, the Lagrangian is a Lorentz scalar and write its transformation as

$$\mathcal{L}'(x') = \mathcal{L}(x). \tag{109}$$

Then, the variation reads

$$\begin{aligned}
\delta\mathcal{L} &= \mathcal{L}'(x) - \mathcal{L}(x) &(110)\\
&= \mathcal{L}(x^\mu - \omega^\mu{}_\nu x^\nu) - \mathcal{L}(x^\mu) &(111)\\
&= -\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}, &(112)
\end{aligned}$$

Nice! Could have done this in 2 and 3

where in the last step we Taylor expanded the term. Now we can write this variation as a total derivative:

$$\delta\mathcal{L} = \partial_\mu(-\omega^\mu{}_\nu x^\nu \mathcal{L}) \implies \Lambda^\mu = -\omega^\mu{}_\nu x^\nu \mathcal{L}, \tag{113}$$

could have argued that $\partial_\mu(\omega^\mu{}_\nu x^\nu) = 0$

and we can apply the Noether's Theorem to find the conserved current:

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)}\delta A_\nu - \Lambda^\mu \tag{114}$$

Explicitly:

$$j^\mu = -F^{\mu\nu}(\omega_{\nu\beta}A^\beta - \omega^\rho{}_\beta x^\beta \partial_\rho A_\nu) + \mathcal{L}\omega^\mu{}_\nu x^\nu. \tag{115}$$

Using the antisymmetry property of $\omega_{\alpha\beta}$ we can reorganize the current identifying the energy momentum tensor from the second question in the same we did on the previous question:

$$\begin{aligned}
j^\mu &= -F^{\mu\nu}(\omega_{\nu\beta}A^\beta - \omega^\rho{}_\beta x^\beta \partial_\rho A_\nu) + \mathcal{L}\omega^\mu{}_\nu x^\nu &(116)\\
&= -F^{\mu\nu}(\omega_{\nu\beta}A^\beta - \omega_{\alpha\beta} g^{\alpha\rho} x^\beta \partial_\rho A_\nu) + \mathcal{L}\omega_{\alpha\nu} g^{\alpha\mu} x^\nu &(117)\\
&= -F^{\mu\nu}(\omega_{\nu\beta}A^\beta - \omega_{\alpha\beta} x^\beta \partial^\alpha A_\nu) + \mathcal{L}\omega_{\alpha\nu} g^{\alpha\mu} x^\nu &(118)\\
&= -F^{\mu\alpha}\omega_{\alpha\beta}A^\beta + F^{\mu\nu}\omega_{\alpha\beta} x^\beta \partial^\alpha A_\nu + \mathcal{L}\omega_{\alpha\beta} g^{\alpha\mu} x^\beta &(119)\\
&= \omega_{\alpha\beta}(F^{\mu\nu} x^\beta \partial^\alpha A_\nu + \mathcal{L} g^{\mu\alpha} x^\beta - F^{\mu\alpha}A^\beta) &(120)\\
&= \omega_{\alpha\beta}(-x^\beta T^{\mu\alpha} - F^{\mu\alpha}A^\beta) &(121)\\
&= \frac{1}{2}\omega_{\alpha\beta}\left[(x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha}) - (F^{\mu\alpha}A^\beta - F^{\mu\beta}A^\alpha)\right]. &(122)
\end{aligned}$$

Finally, we define

$$M^{\alpha\beta\gamma} \equiv x^\alpha T^{\beta\gamma} - x^\gamma T^{\beta\alpha}, \tag{123}$$
$$S^{\alpha\beta\gamma} \equiv F^{\alpha\beta}A^\gamma - F^{\alpha\gamma}A^\beta, \tag{124}$$

very good!

and we write

$$\boxed{j^\mu = \frac{1}{2}\omega_{\alpha\beta}[M^{\alpha\mu\beta} + S^{\mu\beta\alpha}]} \tag{125}$$

→ again, the conserved current is $M^{\mu\alpha\beta} = x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha} + S^{\mu\alpha\beta}$

where M is associated with the orbital angular momentum, and S is associated with spin angular momentum.

The conserved quantities Q are simply obtained by integrating

$$Q = \int d^3x\, j^0. \tag{126}$$

it cannot depend on ω.

Note that it satisfies

$$\partial_t Q = \int d^3x\, \partial_t j^0, \tag{127}$$

but, since j^μ is a conserved current,

$$\partial_\mu j^\mu = 0 \implies \partial_t j^0 = -\nabla \cdot \vec{j}. \tag{128}$$

Therefore,

$$\partial_t Q = -\int d^3x \, \nabla \cdot \vec{j} = -\oint d\vec{S} \cdot \vec{j} = 0, \tag{129}$$

where in the last step we have assumed $\vec{j}$ vanishes at the spatial boundary, since the system is localized. So we have the following conserved quantities:

$$Q^{\alpha\beta} = \int d^3x \, [M^{\alpha 0 \beta} + S^{0\beta\alpha}] = \int d^3x \, [x^\alpha T^{0\beta} - x^\beta T^{0\alpha} + F^{0\beta} A^\alpha - F^{0\alpha} A^\beta]. \tag{130}$$

For $\alpha = i$, $\beta = j$ (spatial components), we obtain the conservation of angular momentum (orbital + spin). For $\alpha = 0$, $\beta = i$ we obtain conserved quantities related to boosts.
